Green Dueling Bandits

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***Abstract*—The dueling bandit model has been acknowledged as an efficient analytic tool for sequential decision-making problems with *qualitative pairwise comparison*. For example, the comparison of workers' completion quality of assigned tasks in crowdsourcing systems; user's ranking of recommended items in recommender systems. In dueling bandits, an agent uses pairwise comparisons of selected arms to balance the exploitation-exploration trade-off during the online learning of uncertainties. Despite the wide application of dueling bandits, their *green* implementation should also consider the non-neglectable *energy costs* for selecting arms, implying the *green dueling bandit* model. Particularly, it requires online control to optimize energy costs adaptively in the long run for sustainable system deployment. Therefore, we 1) employ *online learning* methods to learn the uncertainties via qualitative pairwise comparisons; 2) utilize *online control* techniques to guarantee a within-budget energy cost for the green real-world deployment. Accordingly, we propose a *Green Dueling Bandit Learning (GDBL)* algorithm to effectively integrate dueling bandit learning for the exploration-exploitation trade-off and online control for the optimization of energy costs. We prove that GDBL achieves a sublinear round-averaged regret while keeping the energy cost under budget. We conduct simulations to demonstrate the outperformance of GDBL over baselines.**

I. Introduction

Dueling bandits have gained widespread popularity due to their broad applicability to real-world systems, *e.g.*, crowdsourcing, recommender, and information retrieval systems [1], [2]. As a practical online learning model, dueling bandit handles problems where the quantitive reward is hard to obtain while the *qualitative pairwise comparison* is adopted as feedback during online learning of uncertainties. Take task assignment for crowdsourcing systems (*e.g.*, crowdsourced image captioning) as concrete example (refer to Fig. 1). A crowdsourcing platform (the agent) wants to assign captioning tasks to reliable workers (arms) without prior knowledge of their reliability for tasks. In this case, an absolute relevance score of each candidate caption from selected workers (quantitive reward) is hard to obtain as it requires domain experts to set up detailed evaluation metrics beforehand. Therefore, the platform would estimate such reliability by comparing which candidate caption from workers is more relevant to the given image (qualitative pairwise comparison) across rounds.

Despite dueling bandits' wide applications, their green implementation also needs to consider the *energy costs* incurred by selecting arms. For example, in the mobile crowdsourcing system for data collection, the selected workers (arms) need

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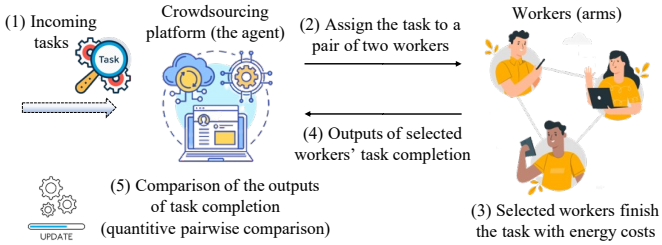


Fig. 1: An illustration of the application scenario of green dueling bandits, *i.e.*, task assignment for crowdsourcing systems.

to use their mobile phones' sensors (*e.g.*, WiFi and GPS) to collect various data and then upload them to the crowdsourcing system. The energy cost of using sensors puts a non-neglectable energy burden on the already limited on-device battery [12]. For green implementation and energy efficiency, we introduce a general *budget constraint* to keep the total energy cost under budget when selecting arms. Notably, such a budget constraint is often imposed in a *long-term* sense for practical systems due to their long-term system deployment with a sustainable green guarantee. For example, both the crowdsourcing platform and the recommender system are deployed to operate for a long time for commercial purposes. Facing such a long-term budget constraint, we employ *online control* techniques to adaptively optimize the incurred energy costs and achieve a within-budget energy cost in the long run.

As argued above, two essential factors are 1) the qualitative pairwise comparison as feedback information in dueling bandits and 2) the long-term budget constraint to optimize the energy cost for green real-world deployment. Due to the missing joint consideration of both factors, we are the first to study them in a *green dueling bandit* model. The comparison between our model and those from existing works is shown in Table I.

Under this new bandit model we face the problem *"How to simultaneously minimize the regret with qualitative pairwise comparisons and guarantee the long-term budget constraint for green implementation?"* Such a problem is challenging as it involves a non-trivial interplay between online learning for regret minimization and online control for constraint satisfaction. On one hand, online control would be misguided with inaccurate estimations if online learning is inefficiently performed. On the other hand, the learning efficiency would be degraded with inferior online feedback if the online control procedure is wrongly conducted. To address the above challenge, we propose

TABLE I: Comparison between Our Green Dueling Bandits and Models from Related Works.

Models	Bandit Settings		Feedback Information		Long-Term Constraint
	Combinatorial MAB	Dueling Bandits	Reward	Pairwise Comparison	
Ours		✓		✓	✓
[3]–[7]		✓		✓	
[8], [9]	✓		✓		✓
[10], [11]	✓		✓		

a *Green Dueling Bandit Learning (GDBL)* algorithm seeking the effective integration of dueling bandit learning and online control. Our main contributions are summarized as follows:

- **Modeling**: To the best of our knowledge, we are the first to study dueling bandits under the general long-term budget constraint for green implementation, *i.e.*, the green dueling bandit model.
- **Algorithm design**: With the effective integration of online learning and online control, we propose GDBL to 1) utilize the pairwise comparison to perform online learning of the uncertainties and 2) adopt online control to guarantee the long-term budget constraint.
- **Theoretical analysis**: We show that GDBL guarantees the long-term budget constraint and obtains a regret bound $O(1/V + \sqrt{\frac{K}{K-1}\log T/T})$ given horizon T and K arms.
- **Numerical results**: Through numerical simulations on various settings, we demonstrate the outperformance of GDBL compared with state-of-the-art baselines.

II. System Model & Problem Formulation

We consider a dueling bandit model with one agent and a set of K arms, denoted as set $[K] \triangleq \{1, \ldots, K\}$. Notably, each arm's importance is not evaluated under some quantitive numerical metric (*e.g.*, reward in conventional multi-armed bandits) but in *pairwise comparison* with each other. Specifically, we denote $p_{i,j}$ as the *dueling probability* of arms i and j, *i.e.*, the probability that arm i wins arm j in pairwise comparisons (*i.e.*, duels). We define the *dueling probability matrix* $\boldsymbol{P}$ with its (i, j) entry as $p_{i,j}$, where $p_{i,i} = 0.5, \forall i \in [K]$ and $p_{i,j} + p_{j,i} = 1, \forall i, j \in [K]$. In each round t within horizon T, the agent simultaneously selects a *dueling pair* of two different arms $i(t)$ and $j(t)$ for comparison. We summarize such selection decisions (*i.e.*, which pair of two arms is selected for comparison) of the agent as subset $D(t) \triangleq \{i(t), j(t)\}, i(t) \neq j(t)$ with $i(t), j(t) \in [K]$. Therefore, we have:

$$D(t) \subset [K], |D(t)| = 2, \tag{1}$$

where $|\cdot|$ denotes the cardinality of a given subset. The agent then obtains the comparison output (*i.e., dueling output*), which follows a Bernoulli distribution parametrized by the dueling probability $p_{i(t),j(t)}$. We denote the dueling output as a random variable $O(t) \sim \text{Bern}(p_{i(t),j(t)})$. Specifically, $O(t) = 1$ means arm i *wins* arm j in the duel of round t and $O(t) = 0$ otherwise.

Definition for the best arm and the different types of winners. We define that arm i *beats* arm j if $q_{i,j} > 1/2$ (denoted as $i \succ j$), *i.e.*, if the probability of winning in the pairwise comparison is larger for arm i than it is for arm j, equivalently, it implies $q_{i,j} > q_{j,i}$. Naturally, one would define the best arm as the arm i such that $p_{i,j} > 1/2, \forall j \neq i$ (referred as the *Condorcet winner* in dueling bandits literature). Despite of being natural, such a definition comes with the major drawback that a Condorcet winner is not guaranteed to exist for every dueling probability matrix. As shown in Fig. 2, we may run into some problem setting, where we cannot decide an overall beat arm with a particular dueling probability matrix. To address this concern, there are different alternative winners (which are all guaranteed to exist for any dueling probability matrix [13]) in the field of dueling bandits to choose from as the definition for the beat arm:

1) *Borda winner* is an arm with the highest *Borda score*. Borda score of arm $i \in [K]$ is defined as

$$\theta_i \triangleq \frac{1}{K-1} \sum\nolimits_{j \neq i} p_{i,j}. \tag{2}$$

For example of $\boldsymbol{P}_2$ in Fig. 2, the corresponding Borda scores are $0.45, 0.55, 0.5$. Intuitively, the Borda score of an arm represents the arm's average probability of winning others. Therefore, the Borda winner has the highest probability of winning an arm selected uniformly and randomly from the remaining arms.

2) *Copeland winner* is an arm with the highest *Copeland score*. Copeland score of arm $i \in [K]$ is defined as

$$\frac{1}{K-1} \sum\nolimits_{j \neq i} \mathbb{I}\left\{p_{i,j} > 1/2\right\},$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function. For example of $\boldsymbol{P}_2$ in Fig. 2, the corresponding Copeland scores are $0.5, 0.5, 0.5$. Note that the Copeland score characterizes the number of arms that an arm can win on average. The Copeland winner is the arm that wins the most number of other arms on average.

3) *von Neumann winner* is a distribution over the arms such that arms sampled from such a distribution on average win against arms sampled from any other distribution. Formally, von Neumann winner is the vector

$$\boldsymbol{w} \in \arg\max_{\boldsymbol{w} \in \Delta_K} \min_{\boldsymbol{u} \in \Delta_K} \boldsymbol{w}^\top \left(\boldsymbol{P} - \frac{1}{2}\boldsymbol{I}\right) \boldsymbol{u},$$

where $\boldsymbol{I}$ denotes the identity matrix and Δ_K denotes the simplex of vectors in $[0, 1]^K$ whose entries sum to 1.

Refer to Fig. 2 for different types of winners in a concrete example. In this paper, we adopt the Borda winner as the definition for the best arm given the following reasons. 1) Different from the Borda score, the Copeland score doesn't directly measure the probability for an arm of winning against

$$\boldsymbol{P}_1 = \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0.5 & 0.6 & 0.8 \\ 0.4 & 0.5 & 0.7 \\ 0.2 & 0.3 & 0.5 \end{bmatrix} \end{array} \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \qquad \boldsymbol{P}_2 = \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0.5 & 0.6 & 0.3 \\ 0.4 & 0.5 & 0.7 \\ 0.7 & 0.3 & 0.5 \end{bmatrix} \end{array} \begin{array}{c} 1 \\ 2 \\ 3 \end{array}$$

Types of Winners in Dueling Bandits	Winner under $\boldsymbol{P}_2$
Condorcet, arm i with $p_{i,j} > \frac{1}{2}, \forall j \neq i$	Non-existence
Borda, $\arg\max_{i\in[K]} \frac{1}{K-1} \sum_{j\neq i} p_{i,j}$	Arm 2
Copeland, $\arg\max_{i\in[K]} \frac{1}{K-1} \sum_{j\neq i} \mathbb{I}\{p_{i,j} > \frac{1}{2}\}$	Arm $i \in \{1,2,3\}$
von Neumann, $\arg\max_{\boldsymbol{w}\in\Delta_K} \min_{\boldsymbol{u}\in\Delta_K} \boldsymbol{w}^\top \left(\boldsymbol{P} - \frac{1}{2}\boldsymbol{I}\right) \boldsymbol{u}$	Distribution $[0.4, 0.4, 0.2]$

Fig. 2: An illustration of three-armed dueling bandits with dueling probability matrices $\boldsymbol{P}_1$ and $\boldsymbol{P}_2$, respectively. The named dots stand for the corresponding arms and the arrows in-between stand for their qualitative relationships, *e.g.*, the arrow from dot 1 to dot 2 means that arm 1 beats arm 2, *i.e.*, $1 \succ 2$. In the case of $\boldsymbol{P}_1$, the overall Condorcet winner exists (*i.e.*, the arm 1); in the case of $\boldsymbol{P}_2$, the Condorcet winner does not exist. The table on the right presents four winners (*i.e.*, the Condorcet, Borda, Copeland, and von Neumann winner) in the case of $\boldsymbol{P}_2$, where $\boldsymbol{I} \in \mathbb{R}^{K\times K}$ denotes the identity matrix and Δ_K denotes the simplex of vectors in $[0,1]^K$ whose entries sum to 1.

others on average like the Borda score does. Therefore, the Copeland winner might lose to others on average, which might be undesirable in practice. 2) Compared with the Borda winner, computing the von Neumann winner (*i.e.*, a distribution over arms) requires more computational, memory, and energy costs in practice, especially when the number of arms is large (*e.g.*, large number of online workers in crowdsourcing systems and larger amount of items in the recommender systems).

Long-term budget constraint for green implementation. We focus on a guarantee of the green implementation of our bandit model in practice. Toward such a goal, we ensure a general within-budget energy cost of selecting arms across rounds. Formally, we denote $C_k(t) \in (0, C_{\max}], \forall k \in [K], \forall t \in [T]$ as the random *energy cost* of selecting arm k in round t, which is assumed known to the agent at the beginning of each round. Therefore, the following budget constraint in (3) is imposed to guarantee the cost is under the budget b ($b > 0$).

$$\limsup_{T'\to\infty} \frac{1}{T'} \sum\nolimits_{t=1}^{T'} \mathbb{E}\left[\sum\nolimits_{k\in D(t)} C_k(t)\right] \leq b. \tag{3}$$

Remark 1. *The reason to consider the long-term budget constraint in* (3) *is that real systems usually run for a long time, and the horizon T is unfixed or unknown in advance. By adopting the "lim" with infinite rounds, we aim to ensure the constraint in a long run without knowing the fixed round T. In this paper, guaranteeing such a long-term budget constraint in* (3) *is equivalently transformed to guarantee the stability of some virtual queue. More details are shown in Section III.*

Problem formulation. We define the *round-averaged regret* $\mathrm{Reg}(T) \triangleq \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[r(t)]$ with the *per-round regret* $r(t)$ defined as $r(t) \triangleq \Theta^* - \frac{1}{2}\sum_{k\in D(t)} \theta_k$, where we denote Θ^* as the Borda score of the optimal dueling pair $D^*(t)$ with $\Theta^* \triangleq \frac{1}{2}\sum_{k\in D^*(t)} \theta_k$. Therefore, we have the problem:

$$\begin{aligned} &\underset{\{D(t)\}_t}{\text{minimize}} \ \mathrm{Reg}(T) \\ &\text{subject to (1), (3).} \end{aligned} \tag{4}$$

Remark 2. *Given the absence of a quantitive performance measure of each arm (i.e., reward in conventional bandit settings) under the dueling bandit formulation, we regard the Borda score in* (2) *as an alternative performance measure. Accordingly, our objective is to minimize the loss of the total Borda score (i.e., the regret). Intuitively, such a minimization process will lead to a dueling pair with the highest Borda score. As a higher Borda score indicates a more important arm that has more chances to win against others, we can identify the beat arm via solving problem* (4).

III. Algorithm Design

In this section, we first present our GDBL algorithm with its online learning and online control components. We then demonstrate the effective integration of learning and control.

Online learning with upper confidence bound. In pursuit of effective online bandit learning of the uncertainties, we estimate the Borda score of each arm $i \in [K]$ as follows:

$$\widehat{\theta}_i(t) \triangleq \frac{1}{K-1} \sum\nolimits_{j\neq i} \widehat{p}_{i,j}(t). \tag{5}$$

We define the *empirical dueling probability* $\widehat{p}_{i,j}(t), i, j \in [K]$ with the Upper Confidence Bound (UCB) estimation as

$$\widehat{p}_{i,j}(t) \triangleq \left[\frac{W_{i,j}(t)}{W_{i,j}(t) + W_{j,i}(t)} + \sqrt{\frac{\alpha \log t}{W_{i,j}(t) + W_{j,i}(t)}}\right]^-, \tag{6}$$

where we denote $[\cdot]^- \triangleq \min\{\cdot, 1\}$ and $W_{i,j}(t)$ as the number of rounds when arm i wins arm j before round t. Note that α is a positive tunable parameter to balance the exploitation-exploration trade-off while estimating the dueling probability. Specifically, the term $\frac{W_{i,j}(t)}{W_{i,j}(t)+W_{j,i}(t)}$ reflects the acquired knowledge about the true dueling probability $p_{i,j}$, which is known as the *exploitation* term; the term $\sqrt{\frac{\log t}{W_{i,j}(t)+W_{j,i}(t)}}$ is the confidence radius. It represents the measurement of how the empirical estimation is close to the ground truth. Specifically, a larger confidence radius implies more uncertainty in the estimation of inter-arm dueling probability, *i.e.*, the arm pair is under-explored with insufficient samples for estimation. Therefore, the confidence radius is also called the *exploration* term, indicating how much an arm pair should be explored.

Remark 3. *Note that the design of estimation $\widehat{p}_{i,j}(t)$ follows the form of the classical UCB1. To effectively balance*

Algorithm 1 Green Dueling Bandit Learning (GDBL)

Initialization: Choose each $\{i,j\}, \forall i \neq j$ of $\binom{K}{2}$ possible dueling pairs once to initialize $W_{i,j}(1)$ and $W_{j,i}(1)$, $\forall i \neq j$; Set the virtual queue length $Q(1) \leftarrow 0$.
for $t = 1$ to T **do**
Compute estimates of Borda score of each arm

$$\widehat{\theta}_i(t) \leftarrow \frac{1}{K-1} \sum_{j \neq i} \widehat{p}_{i,j}(t), i \in [K].$$

Select dueling pair $D(t)$ (breaking a tie randomly)

$$D(t) \in \underset{D \subset [K], |D|=2}{\arg\max} \sum\nolimits_{k \in D} V\widehat{\theta}_k(t)/2 - Q(t)C_k(t).$$

Observe the dueling output $O(t)$.
Update $W_{i,j}(t+1)$ and $W_{j,i}(t+1), \forall i \neq j$.
Update queue length $Q(t+1)$ according to (7).
end for

the exploration-exploitation trade-off, one can also resort to other forms of estimation using methods like MOSS (Minimax Optimal Strategy in the Stochastic case) and TS (Thompson Sampling) [14]. We investigate them numerically in Section V.

Online control with the virtual queue. To guarantee the long-term budget constraint (3), we adopt *Lyapunov optimization with the virtual queue* [15]. Specifically, we introduce a virtual queue Q. In round t, its input is the random cost of selected arms in the dueling pair $D(t)$; and its output is the prescribed budget. Formally, we have:

$$Q(t+1) \triangleq \left[Q(t) + \sum\nolimits_{k \in D(t)} C_k(t) - b \right]^+, \tag{7}$$

where $Q(t)$ is the queue length at the beginning of round t and we adopt the $[\cdot]^+ \triangleq \max\{\cdot, 0\}$ to ensure the non-negative queue length. With such a virtual queue and its dynamics through the arm selection procedure, the budget constraint is guaranteed when the virtual queue is stable, *i.e.*, the queue length $\sum_{t=1}^{T'} Q(t)$ is finite in a round-averaged sense: $\limsup_{T' \to \infty} \frac{1}{T'} \sum_{t=1}^{T'} \mathbb{E}\big[Q(t)\big] < \infty$. Intuitively, this means that we should avoid overloading the virtual queue Q. In other words, we should not frequently select arms of high energy cost to participate in the dueling pair, thus the corresponding queue length would decrease.

Integrated algorithm design. To take into consideration of both regret minimization and guaranteeing the virtual queue stability, we define the *drift-plus-regret* term

$$\Delta_V(Q(t)) \triangleq \mathbb{E}[L(Q(t+1)) - L(Q(t)) + Vr(t)|Q(t)],$$

where the Lyapunov function $L(Q(t)) \triangleq (Q(t))^2/2$ and V is a positive parameter. Intuitively, when the drift-plus-regret term is small, both the variation of queue length and per-round regret are small. This implies bounded queue length and small round-averaged regret. Therefore, our algorithm design seeks an approach to minimize the drift-plus-regret term.

With Lyapunov-drift, we focus on an *upper bound of drift-plus-regret term* (where constant $H_1 \triangleq b^2/2 + 2C_{\max}^2$):

$$\begin{aligned}\Delta_V(Q(t)) \leq\ & H_1 + V\Theta^* + V - Q(t)b \\ & - \sum\nolimits_{k \in D(t)} \mathbb{E}\left[V\widehat{\theta}_k(t)/2 - Q(t)C_k(t) | Q(t) \right]\end{aligned}$$

and minimize such an upper bound in each round t by solving the following per-round problem (8)[1].

$$\begin{aligned}&\underset{D(t)}{\text{maximize}} \sum\nolimits_{k \in D(t)} V\widehat{\theta}_k(t)/2 - Q(t)C_k(t) \\ &\text{subject to } D(t) \subset [K], |D(t)| = 2.\end{aligned} \tag{8}$$

Therefore, our algorithm GDBL (Refer to **Algorithm 1** for its pseudo-code) selects a dueling pair $D(t)$ as

$$D(t) \in \underset{D \subset [K], |D|=2}{\arg\max} \sum\nolimits_{k \in D} V\widehat{\theta}_k(t)/2 - Q(t)C_k(t). \tag{9}$$

Note that V balances the trade-off between each arm's Borda score estimation and the virtual queue length weighted by each arm's energy cost. Intuitively, a larger V implies a larger weight on regret minimization rather than guaranteeing the budget constraint. Refer to Section IV for related theoretical analysis. Besides, due to the selection rule in (9), GDBL tends to select arms with large estimated Borda scores and small energy costs, which matches the goal to identify the best arm while achieving the green implementation.

IV. Theoretical Analysis

We show our main theoretical results, *i.e.*, the queue stability and round-averaged regret of GDBL. We call a budget b feasible if there exists a policy selecting dueling pairs that can guarantee the long-term budget constraint (3) given the budget b. We define the set of all feasible budgets as the *maximal feasibility region*. Suppose there exists b that lies in the interior of the maximal feasible region, we have the following theorems[2].

Theorem 1. *Given a finite V, GDBL achieves queue stability. Formally, there exists a finite positive constant ϵ such that*

$$\limsup_{T' \to \infty} \frac{1}{T'} \sum\nolimits_{t=1}^{T'} \mathbb{E}\big[Q(t)\big] \leq (2H_1 + V)/(2\epsilon) < \infty,$$

where recall that $H_1 = b^2/2 + 2C_{\max}^2$.

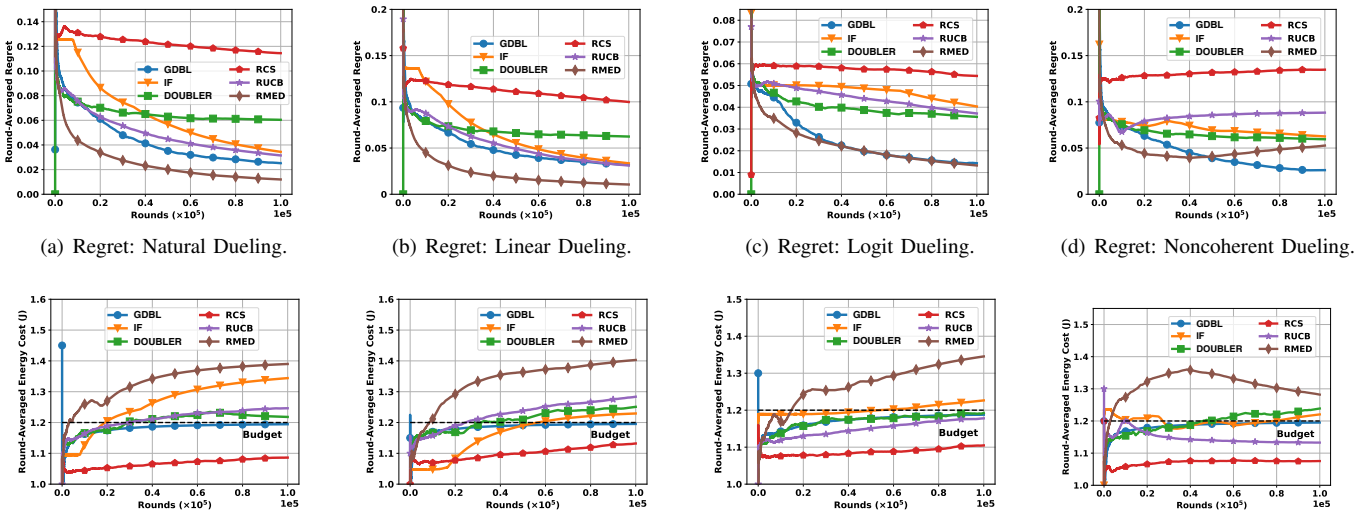
Theorem 2. *GDBL achieves a round-averaged regret*

$$\begin{aligned}\mathrm{Reg}(T) \leq\ & H_1/V + (t_0 + 2H_2)/T \\ & + 2\sqrt{(2\alpha K)/(K-1) \cdot \log T/T},\end{aligned}$$

where $H_2 \triangleq \sum_{t=1}^{\infty} t^{-2\alpha}$; $t_0 \triangleq \binom{K}{2}$ is the rounds for initialization of each possible dueling pair of two arms; parameter $\alpha > 1/2$ to ensure the convergence of the series H_2.

Remark 4. *Theorem 2 shows that the round-averaged regret is at order $O(1/V + \sqrt{\log T/T})$. When $V = \Omega(\sqrt{T/\log T})$, GDBL achieves a sublinear total regret $T \cdot Reg(T)$ at order*

[1]Derivation of the per-round problem is shown in Appendix A.
[2]Proofs of Theorem 1 and 2 are shown in Appendix B and C, respectively.



(a) Regret: Natural Dueling. (b) Regret: Linear Dueling. (c) Regret: Logit Dueling. (d) Regret: Noncoherent Dueling.

(e) Energy: Natural Dueling. (f) Energy: Linear Dueling. (g) Energy: Logit Dueling. (h) Energy: Noncoherent Dueling.

Fig. 3: Performance Comparison between GDBL and the Baselines under Coherent Dueling Modes (*i.e.*, Natural (left), Linear (middle left), and Logit (middle right)) and the Noncoherent Dueling Mode (right).

$O(\sqrt{T\log T})$. *Besides, note that H_1/V in the regret is induced by the online control component. Specifically, to guarantee the long-term budget constraint, GDBL sometimes selects certain high-regret but low-energy arms to lower the total energy costs. Note that such a term is inversely proportional to V. Therefore, with a larger V, GDBL focuses more on minimizing regret rather than guaranteeing the budget constraint, thus a lower regret with a smaller term H_1/V. Therefore, V balances the trade-off between regret minimization and constraint satisfaction, which coincides with our algorithm design in* (9).

V. Simulation Results

We consider a green dueling bandit model of $K = 20$ arms. Recall that we measure the probability of winning or losing for arms in pairwise comparisons using the dueling probability $p_{i,j}$ for arm i and arm j, $\forall i \neq j$. For cases where an overall Condorcet winner exists (denoted as the *coherent dueling*), we model the dueling probability as $p_{i,j} \triangleq \sigma(\mu_i, \mu_j)$, where $\mu_i \in (0, 1)$ denotes the *latent utility* of arm $i, \forall i \in [K]$. By default, such latent utilities are sampled uniformly from interval $[0.2, 0.7]$; the form of function $\sigma(\cdot, \cdot)$ depends on different kinds of *dueling modes*[3]. We consider the following [4]:

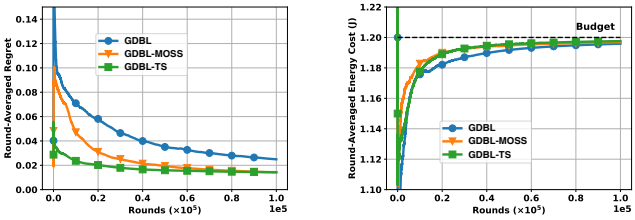
1) *Natural dueling* with $\sigma(\mu_i, \mu_j) = \frac{\mu_i}{\mu_i+\mu_j}$;
2) *Linear dueling* with $\sigma(\mu_i, \mu_j) = (1 + \mu_i - \mu_j)/2$;
3) *Logit dueling* with $\sigma(\mu_i, \mu_j) = (1 + \exp(\mu_j - \mu_i))^{-1}$.

For the case where a Condorcet winner doesn't exist (denoted as the *noncoherent dueling*), we consider a dueling probability matrix as follows: 1) first generate a matrix $\boldsymbol{P}$ following the natural dueling mode under the default setting; 2) set $p_{i,j}, i \in \{1, \ldots, K-1\}, j = i+1$ and $p_{K,1}$ to be larger than 0.5. This implies an ordering for arms with $K \succ 1 \succ 2 \succ \cdots \succ K$. Therefore, there doesn't exist a Condorcet winner. By default, we set the energy cost for each arm in each round as a constant value sampled uniformly from the interval $[0.3J, 0.8J]$, and the budget is set as $b = 1.2J$/round. We set $\alpha = 1$ and $V = 200$ for GDBL and the following experiments run in $T = 10^5$ rounds.

Performance comparison with the baselines. We consider five baselines from the field of dueling bandits in this experiment: 1) *IF*, an explore-then-exploit algorithm [3]; 2) DOUBLER, a UCB-style algorithm that treats dueling bandits as conventional bandits [4]; 3) *RCS*, a two-phase algorithm with round-robin comparisons [6]; 4) *RUCB*, a UCB-style algorithm that estimates all dueling probabilities [5]; 5) *RMED*, an asymptotic optimal algorithm based on the empirical KL divergence [7]. Fig. 3 demonstrates the comparison between GDBL and the baselines under four different dueling modes, *i.e.*, the natural, linear, logit, and noncoherent dueling mode. Under these dueling modes, GDBL effectively guarantees within-budget energy costs while achieving low round-averaged regrets compared with baselines. Since GDBL sometimes has to select high-regret arms for low energy costs, GDBL doesn't achieve the lowest regrets under natural and linear dueling modes in Fig. 3(a) and 3(b). Nevertheless, most baselines cannot keep the energy cost under budget as GDBL can in such dueling modes, thus guarantee of green implementation. Though RCS can achieve lower costs than GDBL in Fig. 3(e), 3(f), 3(g), and 3(h), it cannot effectively balance the regret minimization and the constraint satisfaction, inducing much higher regrets than GDBL in Fig. 3(a), 3(b), 3(c), and 3(d).

Performance comparison with the variants. Recall that

[3]Intuitively, such latency utility measures the quantitive importance of each arm, which is hard to obtain in practice and is never revealed to the agent in our setting. This matches the motivation of using qualitative comparison to perform online learning with dueling bandit methods. Dueling modes correspond to ways of comparing arms with their latent utility. For example, under the natural dueling, arms are compared naturally in terms of their normalized latent utility.



(a) Regret: Natural Dueling. (b) Energy: Natural Dueling.

Fig. 4: Performance Comparison of GDBL and its Variants.

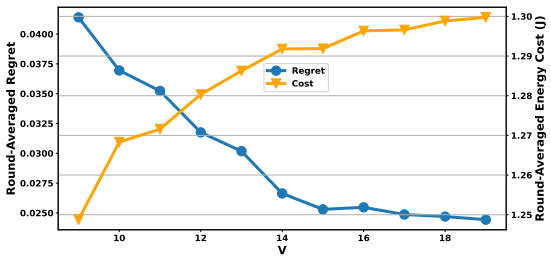


Fig. 5: Effect of Parameter V for GDBL.

GDBL integrates online learning and online control. In the following we consider two variants of GDBL by replacing its online learning component with other alternatives to estimate the dueling probability $p_{i,j}$ with $\widehat{p}_{i,j}(t)$ in (6): 1) *GDBL-MOSS* adopts a MOSS-style estimation with

$$\widehat{p}_{i,j}(t) = \left[\frac{W_{i,j}(t)}{n_{i,j}(t)} + \sqrt{\frac{1}{n_{i,j}(t)}\log^{+}\left(\frac{t}{K\cdot n_{i,j}(t)}\right)}\right]^{-},$$

where recall $[\cdot]^{-} = \min\{\cdot, 1\}$; we define $n_{i,j}(t) \triangleq W_{i,j}(t) + W_{j,i}(t)$ and $\log^{+}(\cdot) \triangleq \log(\max\{\cdot, 1\})$. 2) *GDBL-TS* adopts a TS (Thompson Sampling)-style estimation with

$$\widehat{p}_{i,j}(t) \sim \text{Beta}\big(1 + W_{i,j}(t), 1 + W_{j,i}(t)\big).$$

Therefore, $\widehat{p}_{i,j}(t)$ is a sample from a Beta distribution whose shape parameters are the number of wins ($W_{i,j}(t)$) and losses ($W_{j,i}(t)$) of arm i in duels with arm j before round t.

Fig. 4 demonstrates the comparison between GDBL and its variants under the natural dueling mode. Note that all algorithms achieve low round-averaged regrets while guaranteeing the budget constraint. As expected, by replacing the UCB1 estimation in (6) with advanced MOSS and TS estimation, the achieved round-averaged regret decreases, respectively. This shows the generality of GDBL for the real-world green implementation with various choices of online learning methods.

Effect of parameter V. Recall that parameter V in (9) balances the trade-off between regret minimization and constraint satisfaction, *i.e.*, relative importance between online learning and online control. We conduct an experiment to verify this over a green dueling bandit with $K = 5$ arms under the natural dueling mode. The latent utility of each arm is $[0.2, 0.4, 0.5, 0.6, 0.7]$. The energy cost of each arm is $[0.3J, 0.7J, 0.5J, 0.6J, 0.8J]$ with the budget $b = 1.3J$/round. Fig. 5 demonstrates such a trade-off when parameter V ranges in $[9, 20]$. We note that a larger V implies a larger round-averaged energy cost while a smaller round-averaged regret under GDBL. This verifies our analysis of Theorem 2 in Remark 4, where a larger V means a smaller term in the regret.

VI. Conclusion

In this paper, we studied the qualitative pairwise comparison and the general long-term budget constraint in the green dueling bandit model. With the effective integration of dueling bandit learning methods and Lyapunov optimization techniques, we proposed GDBL to minimize the regret with a green guarantee on energy costs. Theoretical analysis showed that GDBL achieves a sublinear round-averaged regret while guaranteeing the long-term budget constraint. Numerical simulations depicted the outperformance of GDBL over state-of-the-art baselines under various settings.

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APPENDIX

A. Algorithm Design

In this section, we show the our algorithm design based on a transformation of the original problem. We first define the Lyapunov function $L(Q(t)) \triangleq \frac{1}{2}(Q(t))^2$. Recall the update rule for the virtual queue Q as follows:

$$Q(t+1) = \left[Q(t) + \sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D(t)\} - b \right]^+.$$

Therefore, we have the following:

$$\begin{aligned}
& L(Q(t+1)) - L(Q(t)) \\
\leq\ & \frac{1}{2}\left[\left(Q(t) + \sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D(t)\} - b \right)^2 - (Q(t))^2 \right] \\
=\ & \frac{1}{2}\left(\sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D(t)\} - b \right)^2 \\
& + Q(t)\left(\sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D(t)\} - b \right) \\
\leq\ & \frac{1}{2}b^2 + 2C_{\max}^2 + Q(t)\left(\sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D(t)\} - b \right).
\end{aligned}$$

For convenience, we define $H_1 \triangleq \frac{1}{2}b^2 + 2C_{\max}^2$.

Recall the per-round regret $r(t) = \Theta^* - \sum_{k\in D(t)} \theta_k/2$ and we define the drift-plus-regret term $\Delta_V(Q(t)) = \mathbb{E}[L(Q(t+1)) - L(Q(t))|Q(t)] + V\mathbb{E}[r(t)|Q(t)]$. Therefore, we have the following:

$$\begin{aligned}
\Delta_V(Q(t)) \leq\ & H_1 + V\Theta^* - \frac{V}{2}\sum_{k\in D(t)} \mathbb{E}[\theta_k|Q(t)] \\
& + Q(t)\mathbb{E}\left[\sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D(t)\} - b|Q(t) \right].
\end{aligned}$$

Since $\theta_k = \widehat{\theta}_k(t) + \theta_k - \widehat{\theta}_k(t) \geq \widehat{\theta}_k(t) - 1$, we have

$$\begin{aligned}
\Delta_V(Q(t)) \leq\ & H_1 + V\Theta^* + V - Q(t)b \\
& - \mathbb{E}\left[\sum_{k\in D(t)} \frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t)|Q(t) \right]
\end{aligned}$$

To minimize the drift-plus-regret term, we approximately minimize its upper bound by solving the following problem

$$\begin{aligned}
& \underset{D(t)}{\text{maximize}} \sum_{k\in D(t)} \frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t) \\
& \text{subject to } D(t) \subset [K], |D(t)| = 2.
\end{aligned} \tag{10}$$

Therefore, the algorithm selects the dueling pair $D(t)$ as

$$D(t) \in \underset{D\subset[K],|D|=2}{\arg\max} \sum_{k\in D} \frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t).$$

B. Queue Stability

In this section, we prove the queue stability under the proposed algorithm. For the sake of clarity, we denote the selection decision under our proposed algorithm as $\widehat{D}(t)$ in round t. As argued in [8], when the resource budget b lies in the interior of the maximal feasible region, there exists a random policy (whose decision is denoted as $D^\epsilon(t)$) such that

$$\mathbb{E}\left[\sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D^\epsilon(t)\} \right] + \epsilon \leq b, t \in \{1, \ldots, T\}.$$

Given such a ϵ-tightness condition, we in the following investigate the bound of the Lyapunov drift $\Delta L(Q(t)) \triangleq \mathbb{E}[L(Q(t+1)) - L(Q(t))|Q(t)]$:

$$\begin{aligned}
\Delta L(Q(t)) \leq\ & H_1 - Q(t)b \\
& + Q(t)\mathbb{E}\left[\sum_{k\in[K]} C_k(t)\mathbb{I}\left\{k \in \widehat{D}(t)\right\}|Q(t) \right] \\
\leq\ & H_1 + \frac{V}{2} - Q(t)b \\
& - \mathbb{E}\left[\sum_{k\in\widehat{D}(t)} \frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t)|Q(t) \right] \\
\overset{(a)}{\leq}\ & H_1 + \frac{V}{2} - Q(t)b \\
& - \mathbb{E}\left[\sum_{k\in D^\epsilon(t)} \frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t)|Q(t) \right] \\
\leq\ & H_1 + \frac{V}{2} + Q(t)\mathbb{E}\left[\sum_{k\in[K]} C_k(t)\mathbb{I}\{k \in D^\epsilon(t)\} - b|Q(t) \right] \\
\leq\ & H_1 + \frac{V}{2} - \epsilon Q(t),
\end{aligned}$$

where (a) holds since $\widehat{D}(t)$ is the optimal solution to problem 10, thus maximizes $\sum_{k\in\widehat{D}(t)} \frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t)$.

Taking expectation at both sides of above inequality and summing over rounds $\{1, \ldots, T'\}$ gives the following:

$$\begin{aligned}
& \mathbb{E}\Big[L(Q(T'+1))\Big] - \mathbb{E}\Big[L(Q(1))\Big] \\
\leq\ & T'H_1 + \frac{T'V}{2} - \epsilon\sum_{t=1}^{T'} \mathbb{E}\Big[Q(t)\Big].
\end{aligned}$$

Since $L(Q(T'+1)) \geq 0$ and $L(Q(1)) = 0$ and by taking the lim-sup as $T' \to \infty$, we have the following

$$\limsup_{T'\to\infty} \frac{1}{T'}\sum_{t=1}^{T'} \mathbb{E}\left[Q(t)\right] \leq \frac{2H_1 + V}{2\epsilon} < \infty,$$

where recall $H_1 = \frac{1}{2}b^2 + 2C_{\max}^2$ and V is a finite positive parameter.

C. Regret Bound

We denote the selection decision under the optimal policy to the original problem as $D^*(t)$ in round t. It follows that

$$\mathbb{E}\left[\sum_{k\in[K]} C_k(t)\mathbb{I}\left\{k\in D^*(t)\right\}\right] \leq b, t\in\{1,\ldots,T\}.$$

Given that, we study the bound of the drift-plus-regret term:

$$\begin{aligned}\Delta_V(Q(t)) \leq\ & H_1 + V\mathbb{E}\Big[\Theta^* - \sum_{k\in\widehat{D}(t)}\frac{\theta_k}{2}|Q(t)\Big] \\ & + Q(t)\mathbb{E}\left[\sum_{k\in[K]} C_k(t)\mathbb{I}\left\{k\in\widehat{D}(t)\right\} - b|Q(t)\right] \\ \leq\ & H_1 + \mathbb{E}\Big[\Phi_1(t)|Q(t)\Big],\end{aligned}$$

where for convenience, we define

$$\begin{aligned}\Phi_1(t) \triangleq\ & \sum_{k\in D^*(t)}\left(\frac{V}{2}\theta_k - Q(t)C_k(t)\right) \\ & - \sum_{k\in\widehat{D}(t)}\left(\frac{V}{2}\theta_k - Q(t)C_k(t)\right).\end{aligned}$$

Therefore, taking expectation and summing over $\{1,\ldots,T\}$ of $\Delta_V(Q(t))$ gives the following:

$$\begin{aligned}& \frac{1}{TV}\Big(\mathbb{E}\Big[L(Q(T+1))\Big] - \mathbb{E}\Big[L(Q(1))\Big]\Big) + \text{Reg}(T) \\ \leq\ & \frac{H_1}{V} + \frac{1}{TV}\sum_{t=0}^{T-1}\mathbb{E}\Big[\Phi_1(t)\Big].\end{aligned}$$

Since $L(Q(T+1))\geq 0$ and $L(Q(1))=0$, we have

$$\text{Reg}(T) \leq \frac{H_1}{V} + \frac{1}{TV}\sum_{t=1}^{T}\mathbb{E}\Big[\Phi_1(t)\Big].$$

We assume a policy that solves problem (11) optimally, whose selection decision is denoted as $D^{'}(t)$ in round t.

$$\underset{D(t)\subset[K],|D(t)|=2}{\text{maximize}} \sum_{k\in D(t)}\frac{V}{2}\theta_k(t) - Q(t)C_k(t) \qquad (11)$$

Therefore, we have the following inequality:

$$\begin{aligned}& \sum_{k\in D^*(t)}\frac{V}{2}\theta_k - Q(t)C_k(t) \\ \leq\ & \sum_{k\in D'(t)}\frac{V}{2}\theta_k - Q(t)C_k(t).\end{aligned}$$

Recall the per-round problem (10), we have:

$$\begin{aligned}& \sum_{k\in D'(t)}\frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t) \\ \leq\ & \sum_{k\in\widehat{D}(t)}\frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t).\end{aligned}$$

Therefore, we the derive a bound of $\Phi_1(t)$ as follows:

$$\begin{aligned}\Phi_1(t) \leq\ & \sum_{k\in D'(t)}\left(\frac{V}{2}\theta_k - Q(t)C_k(t)\right) \\ & - \sum_{k\in\widehat{D}(t)}\left(\frac{V}{2}\theta_k - Q(t)C_k(t)\right) \\ & + \sum_{k\in\widehat{D}(t)}\left(\frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t)\right) \\ & - \sum_{k\in D'(t)}\left(\frac{V}{2}\widehat{\theta}_k(t) - Q(t)C_k(t)\right) \\ =\ & V\sum_{k\in\widehat{D}(t)}\frac{\widehat{\theta}_k(t)-\theta_k}{2} + V\sum_{k\in D'(t)}\frac{\theta_k-\widehat{\theta}_k(t)}{2}.\end{aligned}$$

Recall that

$$\widehat{\theta}_i(t) - \theta_i = \frac{1}{K-1}\sum_{j\neq i}(\widehat{p}_{i,j}(t) - p_{i,j}).$$

Therefore, we have:

$$\Phi_1(t) \leq \frac{V}{K-1}\left(\Phi_2(t) + \Phi_3(t)\right),$$

where we define

$$\begin{aligned}\Phi_2(t) &\triangleq \sum_{i\in\widehat{D}(t)}\sum_{j\neq i}\frac{\widehat{p}_{i,j}(t) - p_{i,j}}{2}, \\ \Phi_3(t) &\triangleq \sum_{i\in D'(t)}\sum_{j\neq i}\frac{p_{i,j} - \widehat{p}_{i,j}(t)}{2}.\end{aligned}$$

In summary, we have

$$\text{Reg}(T) \leq \frac{H_1}{V} + \frac{1}{T(K-1)}\mathbb{E}\left[\sum_{t=1}^{T}\Phi_2(t) + \sum_{t=1}^{T}\Phi_3(t)\right].$$

Bounding $\sum_{t=1}^{T}\mathbb{E}[\Phi_2(t)]$. Recall the initialization process of our proposed algorithm where we spend the first $t_0 = \binom{K}{2}$ rounds to select each dueling pair once. Therefore, we have:

$$\sum_{t=1}^{T}\Phi_2(t) \leq \phi_2 + (K-1)t_0,$$

where we define

$$\phi_2 \triangleq \sum_{t=t_0+1}^{T}\sum_{i\in\widehat{D}(t)}\sum_{j\neq i}\frac{\widehat{p}_{i,j}(t) - p_{i,j}}{2}$$

We define the event (for $i\neq j$)

$$G_{i,j}(t) \triangleq \left\{\frac{W_{i,j}(t)}{W_{i,j}(t)+W_{j,i}(t)} - p_{i,j} > \sqrt{\frac{\alpha\log t}{W_{i,j}(t)+W_{j,i}(t)}}\right\}$$

and let $G_{i,j}^c(t)$ be its complement. When event $G_{i,j}(t)$ happens, it induces $\widehat{p}_{i,j}(t)\geq p_{i,j}$; when event $G_{i,j}^c(t)$ happens, it induces

$\widehat{p}_{i,j}(t) - p_{i,j} \leq 2\sqrt{\frac{\alpha \log t}{W_{i,j}(t)+W_{j,i}(t)}}$ Therefore, we have

$$\begin{aligned}
\phi_2 \leq & \sum_{t=t_0+1}^{T} \sum_{i\in\widehat{D}(t)} \sum_{j\neq i} \frac{\widehat{p}_{i,j}(t) - p_{i,j}}{2} \mathbb{I}\left\{\widehat{p}_{i,j}(t) \geq p_{i,j}\right\} \\
= & \sum_{t=t_0+1}^{T} \sum_{i\in\widehat{D}(t)} \sum_{j\neq i} \frac{\widehat{p}_{i,j}(t) - p_{i,j}}{2} \mathbb{I}\left\{\widehat{p}_{i,j}(t) \geq p_{i,j}\right\} \mathbb{I}\left\{G_{i,j}(t)\right\} \\
& + \sum_{t=t_0+1}^{T} \sum_{i\in\widehat{D}(t)} \sum_{j\neq i} \frac{\widehat{p}_{i,j}(t) - p_{i,j}}{2} \mathbb{I}\left\{\widehat{p}_{i,j}(t) \geq p_{i,j}\right\} \mathbb{I}\left\{G^c_{i,j}(t)\right\} \\
\leq & \sum_{t=t_0+1}^{T} \phi_{2,1} + \sum_{t=t_0+1}^{T} \phi_{2,2},
\end{aligned}$$

where we define

$$\begin{aligned}
\phi_{2,1} &= \sum_{i\in\widehat{D}(t)} \sum_{j\neq i} \frac{\widehat{p}_{i,j}(t) - p_{i,j}}{2} \mathbb{I}\left\{G_{i,j}(t)\right\} \\
\phi_{2,2} &= \sum_{i\in\widehat{D}(t)} \sum_{j\neq i} \sqrt{\frac{\alpha \log t}{W_{i,j}(t) + W_{j,i}(t)}}.
\end{aligned}$$

We define the history as $F(t) \triangleq \{D(\tau), W_{i,j}(\tau)\}_{\tau<t}$. Therefore, we have

$$\begin{aligned}
& \mathbb{E}[\phi_{2,1}|F(t)] \\
\leq & \sum_{i\in[K]} \sum_{j\neq i} \frac{\mathbb{P}\left\{G_{i,j}(t)\right\}}{2} \\
\leq & \ \mathbb{P}\left\{\frac{W_{i,j}(t)}{W_{i,j}(t)+W_{j,i}(t)} - p_{i,j} > \sqrt{\frac{\alpha \log t}{W_{i,j}(t)+W_{j,i}(t)}}\right\} \\
\leq & \ \exp\{-2\alpha \log t\} = t^{-2\alpha}.
\end{aligned}$$

Therefore, by Adam's law, we have

$$\begin{aligned}
& \sum_{t=t_0+1}^{T} \mathbb{E}[\phi_{2,1}(t)] \\
= & \sum_{t=t_0+1}^{T} \mathbb{E}[\mathbb{E}[\phi_{2,1}(t)|F(t)]] \\
\leq & \ (K-1) \sum_{t=t_0+1}^{T} t^{-2\alpha} \\
\leq & \ (K-1) H_2,
\end{aligned}$$

where we define $H_2 \triangleq \sum_{t=1}^{\infty} t^{-2\alpha}$.

Note that

$$\begin{aligned}
& \sum_{i\in[K]} \sum_{j\neq i} \frac{\mathbb{I}\{i \in \widehat{D}(t)\}}{\sqrt{W_{i,j}(t) + W_{j,i}(t)}} \\
\leq & \sum_{i<j} \sum_{n=2}^{W_{i,j}(T+1)+W_{j,i}(T+1)} \frac{2}{\sqrt{n-1}} \\
\leq & \ 2\sum\nolimits_{i<j} \int_1^{W_{i,j}(T+1)+W_{j,i}(T+1)} \frac{1}{\sqrt{n}} dn \\
\leq & \ 4\sum\nolimits_{i<j} \sqrt{W_{i,j}(T+1) + W_{j,i}(T+1)} \\
\leq & \ 4\sqrt{\frac{K(K-1)}{2} \sum_{i<j} W_{i,j}(T+1) + W_{j,i}(T+1)} \\
= & \ 2\sqrt{2K(K-1)T}.
\end{aligned}$$

Therefore, we have

$$\begin{aligned}
\sum\nolimits_{t=t_0+1}^{T} \phi_{2,2}(t) \leq & \ \sqrt{\alpha \log T} \sum_{i\in[K]} \sum_{j\neq i} \frac{\mathbb{I}\{i \in \widehat{D}(t)\}}{\sqrt{W_{i,j}(t) + W_{j,i}(t)}} \\
\leq & \ 2\sqrt{2\alpha K(K-1)T \log T}.
\end{aligned}$$

In summary, we have

$$\begin{aligned}
\sum\nolimits_{t=1}^{T} \mathbb{E}[\Phi_2] \leq & \ (K-1)(t_0 + H_2) \\
& + 2\sqrt{2\alpha K(K-1)T \log T}.
\end{aligned}$$

Bounding $\sum_{t=1}^{T} \mathbb{E}[\Phi_3(t)]$. When $p_{i,j} > \widehat{p}_{i,j}(t)$, it induces that $p_{i,j} > \frac{W_{i,j}(t)}{W_{i,j}(t)+W_{j,i}(t)} + \sqrt{\frac{\alpha \log t}{W_{i,j}(t)+W_{j,i}(t)}}$. Then, we have

$$\begin{aligned}
& \mathbb{P}\left\{p_{i,j} > \widehat{p}_{i,j}(t)\right\} \\
\leq & \ \mathbb{P}\left\{p_{i,j} > \frac{W_{i,j}(t)}{W_{i,j}(t)+W_{j,i}(t)} + \sqrt{\frac{\alpha \log t}{W_{i,j}(t)+W_{j,i}(t)}}\right\} \\
\leq & \ t^{-2\alpha}.
\end{aligned}$$

Given that, we have

$$\begin{aligned}
& \sum\nolimits_{t=1}^{T} \mathbb{E}[\Phi_3(t)] \leq (K-1)t_0 \\
& + \sum_{t=t_0+1}^{T} \sum_{i\in\widehat{D}(t)} \sum_{j\neq i} \frac{p_{i,j} - \widehat{p}_{i,j}(t)}{2} \mathbb{P}\left\{p_{i,j} > \widehat{p}_{i,j}(t)\right\} \\
& \leq (K-1)t_0 + \sum_{t=t_0+1}^{T} \sum_{i\in[K]} \sum_{j\neq i} t^{-2\alpha} \frac{\mathbb{I}\{i \in \widehat{D}(t)\}}{2} \\
& \leq (K-1) \sum\nolimits_{t=1}^{\infty} t^{-2\alpha} = (K-1)H_2.
\end{aligned}$$

Finally, we bound the round-averaged regret as follows:

$$\begin{aligned}
\mathrm{Reg}(T) \leq & \ \frac{H_1}{V} + \frac{1}{T(K-1)} \mathbb{E}\left[\sum_{t=1}^{T} \Phi_2(t) + \sum_{t=1}^{T} \Phi_3(t)\right] \\
\leq & \ \frac{H_1}{V} + \frac{t_0 + 2H_2}{T} + 2\sqrt{\frac{2\alpha K}{K-1} \frac{\log T}{T}},
\end{aligned}$$

where $H_1 = \frac{1}{2}b^2 + 2C^2_{\max}$ and $H_2 = \sum_{t=1}^{\infty} t^{-2\alpha}$.